Optimization and Approximation
TD 7: Modelization and Linear Programming

1 Introduction

A *linear program* is a minimization or maximization problem of some linear *objective function* in terms of *decision variables* subject to some linear equalities or inequalities called *constraints*.

Example 1 - Polly's diet (tribute to Vašek Chvátal)

Nutrition Facts
(fictive figures)

	Proteins	Fat	Sugar	Calories
Chicken	1	2	0	4
Egg	1	3	1	5
Bread	2	0	1	2
Milk (liter)	1	1	1	3

Recommended Daily Amounts: At least 10 proteins, 15 fat and 6 sugar.
Not recommended: Exceed 4 chicken a day or 5 liter of milk.
Problem: Design a diet with minimum calories.

We want to write the associate linear program:

(a) Identify the decision variables in this problem.

(b) Express the constraints.

(c) Express the value that we want to optimize in function of the decision variables.

(a) The decision variables are the daily quantities of each food: x_c (chicken), x_e (eggs), x_b (bread) and x_m (liters of milk).

(b) Nutritional requirements, upper limits and non-negativity:

$$\begin{aligned} x_c + x_e + 2x_b + x_m &\geq 10 && \text{(proteins)} \\ 2x_c + 3x_e + x_m &\geq 15 && \text{(fat)} \\ x_e + x_b + x_m &\geq 6 && \text{(sugar)} \\ x_c \leq 4, \quad x_m &\leq 5, \\ x_c, x_e, x_b, x_m &\geq 0. \end{aligned}$$

(c) We minimize the number of calories:

$$\min \ 4x_c + 5x_e + 2x_b + 3x_m.$$

Example 2 - Canonical form

A linear program is in *canonical* or *standard* form if it reads:

$$(P)\quad \begin{array}{ll} \text{Maximise} & \displaystyle\sum_{j=1}^{n} c_j x_j \qquad \text{(usually } n \text{ variables)} \\ \text{Subject to} & \left.\begin{array}{l} \displaystyle\sum_{j=1}^{n} a_{ij} x_j \le b_i, \quad \text{for } i = 1, \dots, m \\ x_1, \dots, x_n \ge 0 \end{array}\right\} \; m+n \text{ constraints.} \end{array}$$

For each of the following changes, write an equivalent linear program in standard form:

(a) Minimize $\sum_{j=1}^{n} c_j x_j$.

(b) $\sum_{j=1}^{n} a_{ij} x_j \ge b_i, \quad \text{for } i = 1, \dots, m$

(c) $\sum_{j=1}^{n} a_{ij} x_j = b_i, \quad \text{for } i = 1, \dots, m$

(d) The decisions variables are not necessarily non-negatives.

(a) Since $\min_x \mathbf{c}^T\mathbf{x} = -\max_x(-\mathbf{c})^T\mathbf{x}$, we maximize $\sum_{j=1}^{n}(-c_j)x_j$ under the same constraints. The optimal solutions are the same and the optimal values are opposite.

(b) Multiply by -1: $\sum_{j=1}^{n}(-a_{ij})x_j \le -b_i$, for $i = 1, \dots, m$.

(c) Replace each equality by two inequalities:

$$\sum_{j=1}^{n} a_{ij} x_j \le b_i \quad \text{and} \quad \sum_{j=1}^{n}(-a_{ij})x_j \le -b_i, \qquad i = 1, \dots, m.$$

(d) Write each free variable as the difference of two non-negative variables: $x_j = x_j^+ - x_j^-$ with $x_j^+, x_j^- \ge 0$, and substitute it in the objective and in the constraints. Every real x_j can be written this way (e.g. $x_j^+ = \max(x_j, 0)$, $x_j^- = \max(-x_j, 0)$), and conversely every pair (x_j^+, x_j^-) gives a real x_j, so the two programs have the same optimal value.

Let (P) be an l.p. An n-tuple $(x_1, \dots, x_n)$ is an *admissible solution* of (P) if it satisfies the constraints. If it moreover maximizes (or minimizes) the objective function it is an *optimal solution*. We call *domain of* (P) the set of admissible solutions.

Create an example for each of the three possible outcomes:

- The domain of (P) is empty, hence no solution. (P) is then said *infeasible*.
- The domain of (P) is unbounded, we either have optimal solutions, or the objective function is unbounded (in the later case (P) is said *unbounded*).
- The domain of (P) is non-empty and bounded, hence there exists an optimal solution.

- *Infeasible:* $\max x_1$ subject to $x_1 \leq -1$, $x_1 \geq 0$. No x_1 satisfies both constraints.
- *Unbounded domain:*
  - without optimal solution: $\max x_1$ subject to $-x_1 \leq 0$, $x_1 \geq 0$. Every $x_1 \geq 0$ is admissible and the objective tends to $+\infty$;
  - with optimal solutions: $\max -x_1 - x_2$ subject to $-x_1 - x_2 \leq -1$, $x_1, x_2 \geq 0$. The domain $\{x_1 + x_2 \geq 1,\ x_1, x_2 \geq 0\}$ is unbounded, but the objective is at most -1, and this value is reached on the whole segment $x_1 + x_2 = 1$, $x_1, x_2 \geq 0$.
- *Bounded domain:* $\max x_1$ subject to $x_1 + x_2 \leq 1$, $x_1, x_2 \geq 0$. The domain is the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$ and the unique optimal solution is $(1,0)$, with value 1.

2 Exercises

Indication of hardness: from (*) to (****).

Exercise 1 - Lumberjack (*)

A lumberjack has 100 hectares of hardwoods. Cutting a hectare of wood and letting the zone regenerate naturally cost 10k$ per hectare and the benefit is 50k$. Alternatively, cutting a hectare of wood and re-seeding with pine-woods costs 50k$ per hectare, and the benefit is 120k$. We assume that the lumberjack has only 4000k$ funds.

What should the lumberjack do to maximise his benefit?

(a) Formulate the problem as a LP.

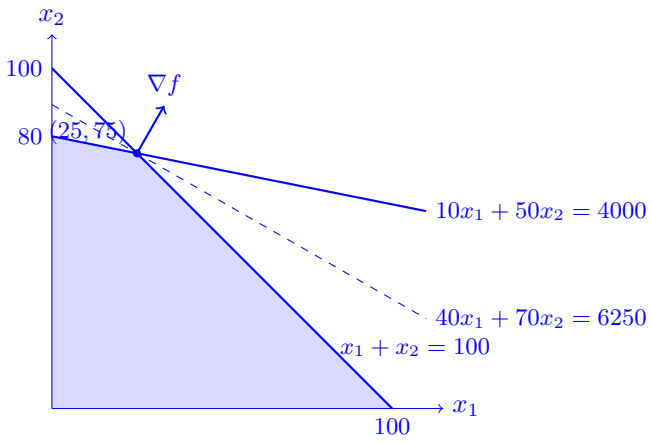
(b) Represent the LP geometrically.

(c) Find the best strategy to adopt and the benefit associated.

(a) Let x_1 be the number of hectares cut and left to regenerate naturally, and x_2 the number of hectares cut and re-seeded with pines. We read "benefit" as the revenue, so the net profit per hectare is $50 - 10 = 40$k$ for the first option and $120 - 50 = 70$k$ for the second. The LP is (in k$)

$$\begin{aligned} \max\ & 40x_1 + 70x_2 & \\ \text{s.t. } & x_1 + x_2 \leq 100 & \text{(area)} \\ & 10x_1 + 50x_2 \leq 4000 & \text{(funds)} \\ & x_1, x_2 \geq 0. & \end{aligned}$$

(b) The domain is the quadrilateral with vertices $(0,0)$, $(100,0)$, $(25,75)$ and $(0,80)$. The vertex $(25,75)$ is the intersection of the two lines $x_1 + x_2 = 100$ and $x_1 + 5x_2 = 400$. The level lines of the objective $40x_1 + 70x_2 = c$ are parallel lines that we push in the direction $\nabla f = (40,70)^T$ while they still meet the domain (dashed line).



(c) The optimum of a bounded LP is reached at a vertex of the domain. Evaluating the objective:

$f(0,0) = 0, \qquad f(100,0) = 4000, \qquad f(0,80) = 5600, \qquad f(25,75) = 1000+5250 = 6250.$

The best strategy is to let 25 hectares regenerate naturally and to re-seed 75 hectares with pines, which uses all the land and all the funds. The profit is 6250k\$. Geometrically, the slope of the level lines ($-40/70 \approx -0.57$) lies between the slopes of the two constraints (-1 and $-1/5$), so the last point touched is their intersection.

Exercise 2 - Independent Set Problem (*)

Given a graph $G = (V, E)$, an independent set is a set of vertices pairwise not adjacent. Formulate the Maximum Independent Set problem as an ILP.

For each vertex $v \in V$ let $x_v \in \{0, 1\}$, with $x_v = 1$ if and only if v belongs to the set. The ILP is

$$\begin{aligned} \max \ & \sum_{v \in V} x_v & \\ \text{s.t. } & x_u + x_v \leq 1 & \forall\, uv \in E, \\ & x_v \in \{0, 1\} & \forall\, v \in V. \end{aligned}$$

The constraint says that the two endpoints of an edge cannot both be chosen. So the admissible solutions are exactly the indicator vectors of independent sets, and the objective is the size of the set.

Exercise 3 - N-queens problem (**)

Suppose that you are given an $N \times N$-chessboard. Formulate the problem of placing N queens on the board such that no two queens share any row, column or diagonal as an Integer Linear Programming problem.

For $i, j \in \{1, \ldots, N\}$ let $x_{ij} \in \{0, 1\}$, with $x_{ij} = 1$ if and only if there is a queen on square

(i, j) (row i, column j). The squares of a diagonal have constant $i - j$, and the squares of an anti-diagonal have constant $i + j$. There are N queens and N rows, and each row holds at most one queen, so each row holds exactly one; the same holds for columns. The ILP is

$$\sum_{j=1}^{N} x_{ij} = 1 \qquad i = 1, \ldots, N \qquad \text{(rows)}$$

$$\sum_{i=1}^{N} x_{ij} = 1 \qquad j = 1, \ldots, N \qquad \text{(columns)}$$

$$\sum_{\substack{1 \le i,j \le N \\ i-j=k}} x_{ij} \le 1 \qquad k = -(N-1), \ldots, N-1 \qquad \text{(diagonals)}$$

$$\sum_{\substack{1 \le i,j \le N \\ i+j=k}} x_{ij} \le 1 \qquad k = 2, \ldots, 2N \qquad \text{(anti-diagonals)}$$

$$x_{ij} \in \{0, 1\} \qquad i, j = 1, \ldots, N.$$

This is a feasibility problem: any objective works, e.g. $\min 0$. Equivalently, we can replace the row and column equalities by ≤ 1 and maximize $\sum_{i,j} x_{ij}$: the placement exists if and only if the optimal value is N.